Incremental Data-Driven Robust Crew Pairing Optimization

Under Evolving Uncertainty

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Background $\rightarrow$ Literature $\rightarrow$ Research gap $\rightarrow$ Objective

Background: Crew Pairing

What is crew pairing?

- Assign crews to sequences of flights.
- Cover every flight exactly once.
- Respect crew duty and rest rules.

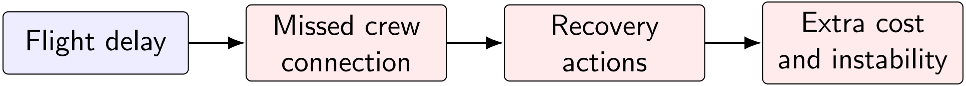
Why is it important?

- Crew cost is a major airline cost.
- Feasible schedules support daily operations.
- Disruptions can quickly affect later flights.

Core planning question

How can an airline design crew pairings that are low-cost and operationally reliable?

Motivation: Delay Uncertainty



- Delay uncertainty can break planned crew connections.
- Recovery may require waiting, swapping, or cancellation.
- Robustness is needed, not only nominal cost minimization.

Related Literature

Robust optimization

- Budgeted uncertainty.
- Robustness-cost trade-off.
- Bertsimas and Sim (2004).

Two-stage methods

- Recourse after uncertainty.
- Column-and-constraint generation.
- Zhao and Zeng (2012); Zeng and Zhao (2013).

Transportation planning

- Robust network and airline recovery.
- Column generation for large pairing spaces.
- Wang and Qi (2020).

Research Gap

1. **Distributional assumptions may be fragile.** Rare delay events can be diluted by many normal observations.
2. **Delay uncertainty is asymmetric.** Late arrivals create operational risk; early arrivals are usually absorbed.
3. **Uncertainty evolves over time.** New data should update the model without rebuilding everything from zero.

Gap addressed by this project

A data-driven robust crew pairing framework with incremental re-optimization under evolving uncertainty.

Research Objective And Direction

Objective

Develop a crew pairing optimization framework that is robust, data-driven, and adaptive to new delay data.

- Use empirical quantiles to construct delay bounds:

$$d_f = \mathrm{Quantile}_\alpha(x_f^1, x_f^2, \ldots, x_f^T)$$

- Use budgeted uncertainty to balance protection and cost.
- Reuse previous solutions and constraints when uncertainty changes.

Expected Contribution

- **Modeling contribution:** quantile-based, one-sided uncertainty set for delay disruptions.
- **Optimization contribution:** two-stage robust crew pairing with recovery actions.
- **Algorithmic contribution:** nested C&CG and column generation for large-scale solution.
- **Practical contribution:** incremental re-optimization as delay information evolves.

Conclusion

- Crew pairing is cost-sensitive and operationally critical.
- Delay uncertainty can make nominal schedules unreliable.
- Related literature motivates robust, two-stage, and decomposition-based methods.
- This project focuses on data-driven uncertainty and incremental model updates.

Final message

The project introduces a robust crew pairing framework that can learn from delay data and adapt as uncertainty evolves.